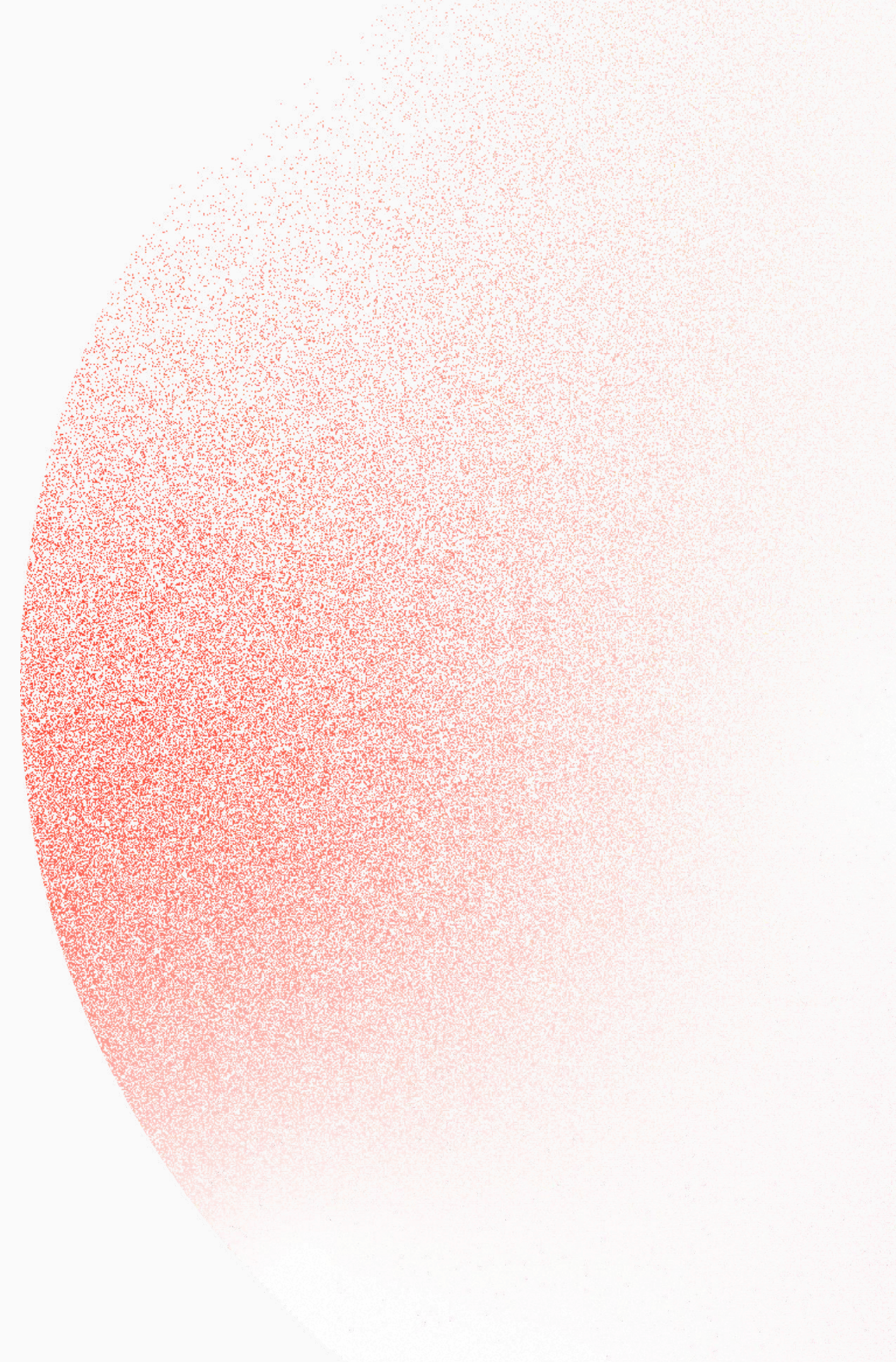


Architecture of Cloud / Network for AI Computing

Arnon Tawong

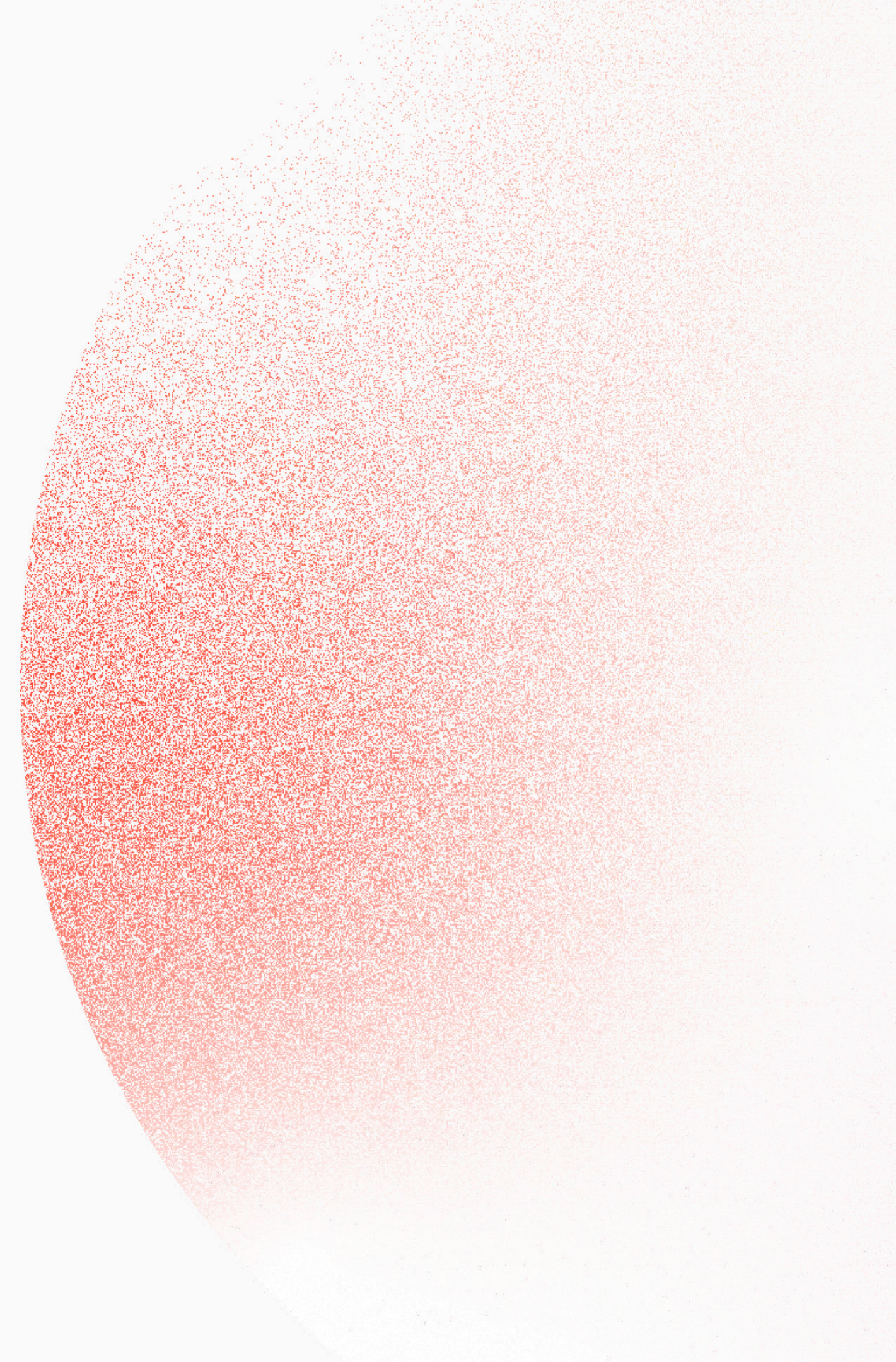
DIGITAL & AI IN HEALTH SYSTEM -
CHULALONGKORN UNIVERSITY

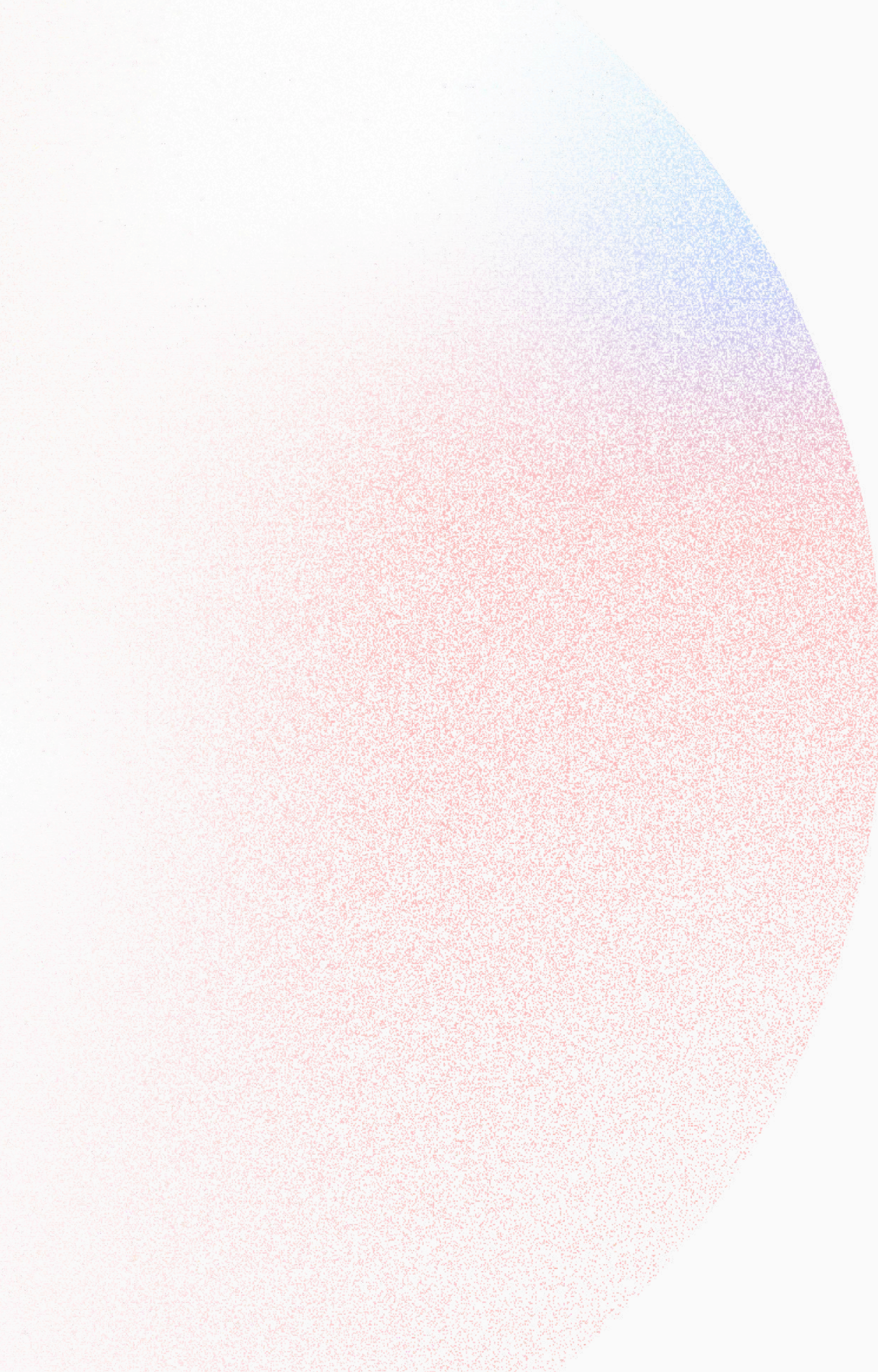


Level

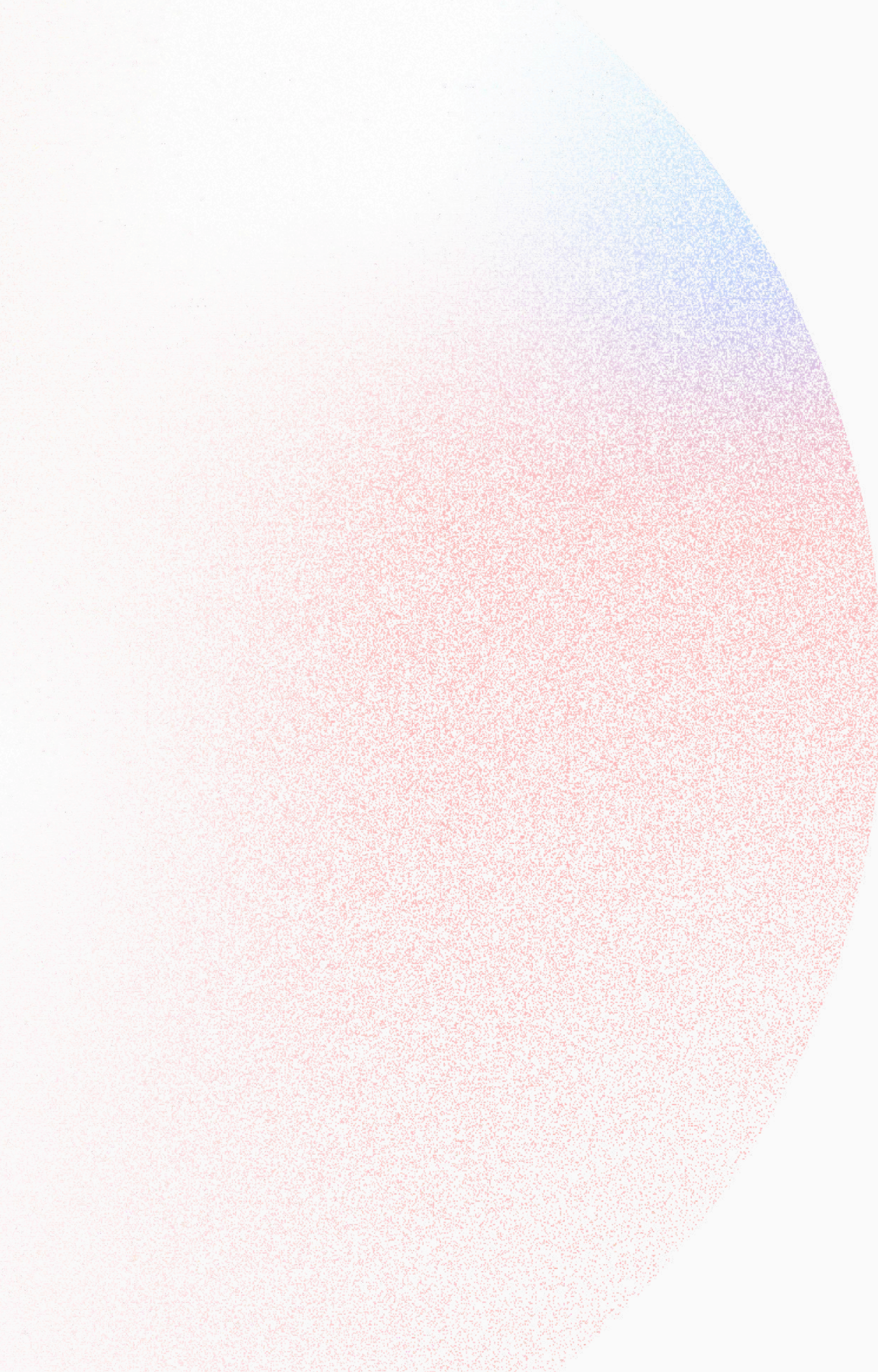


Mostly Jargon and
Systematic Thinking





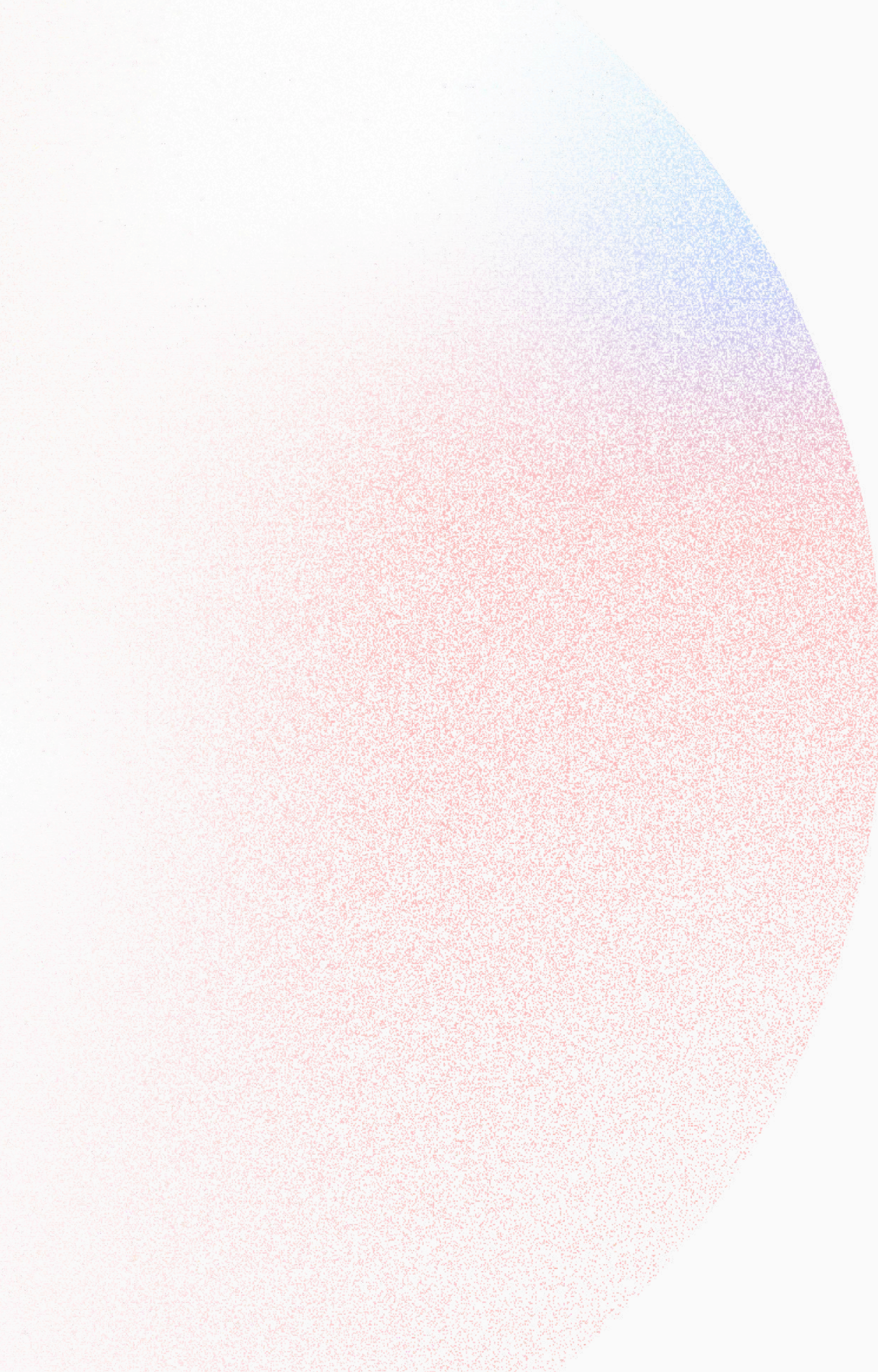
What is Cloud ?



Short Answer

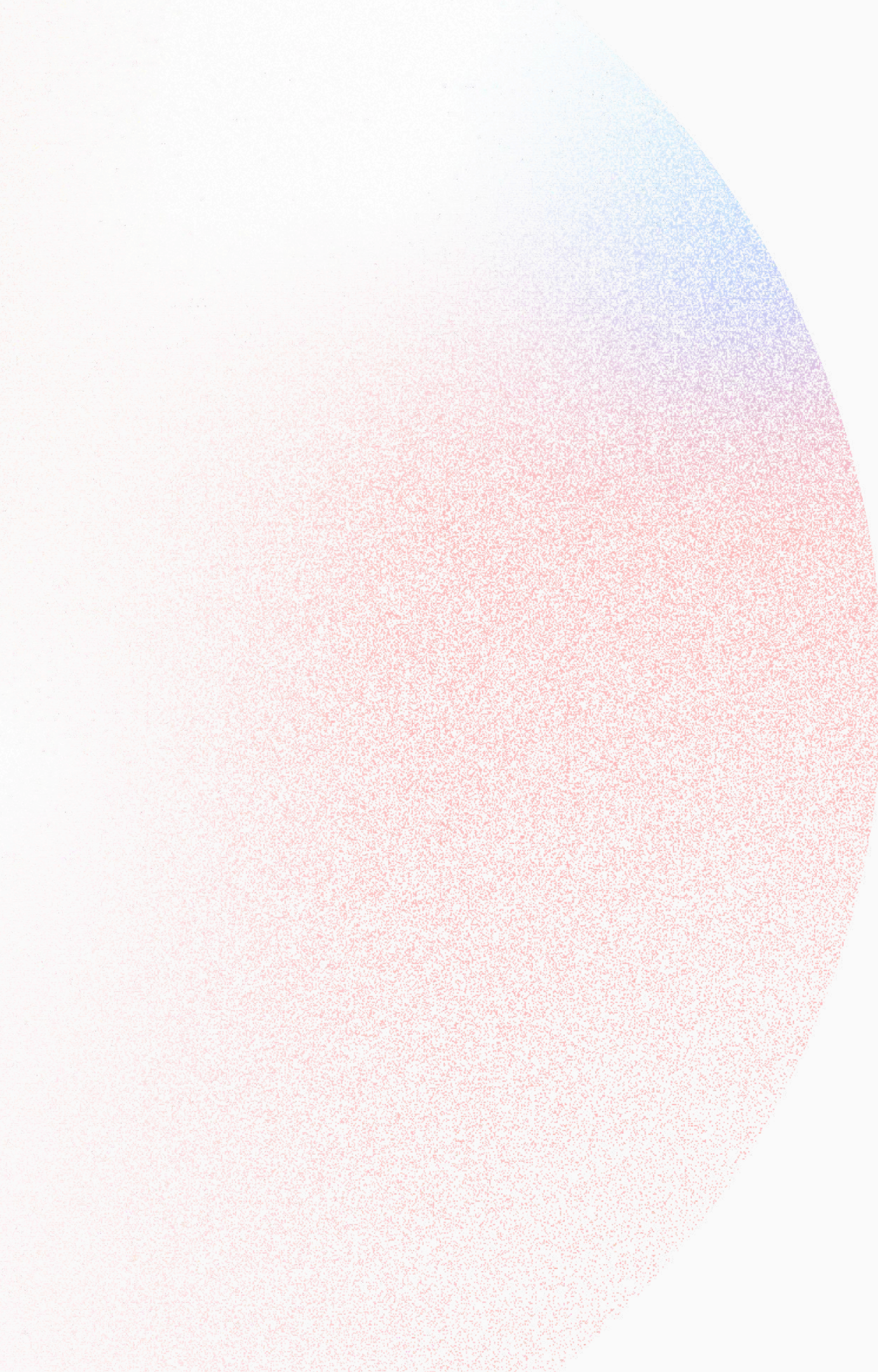
Someone Else Server

Yeah, That it.



What is Serverless ?

Does not mean no server na

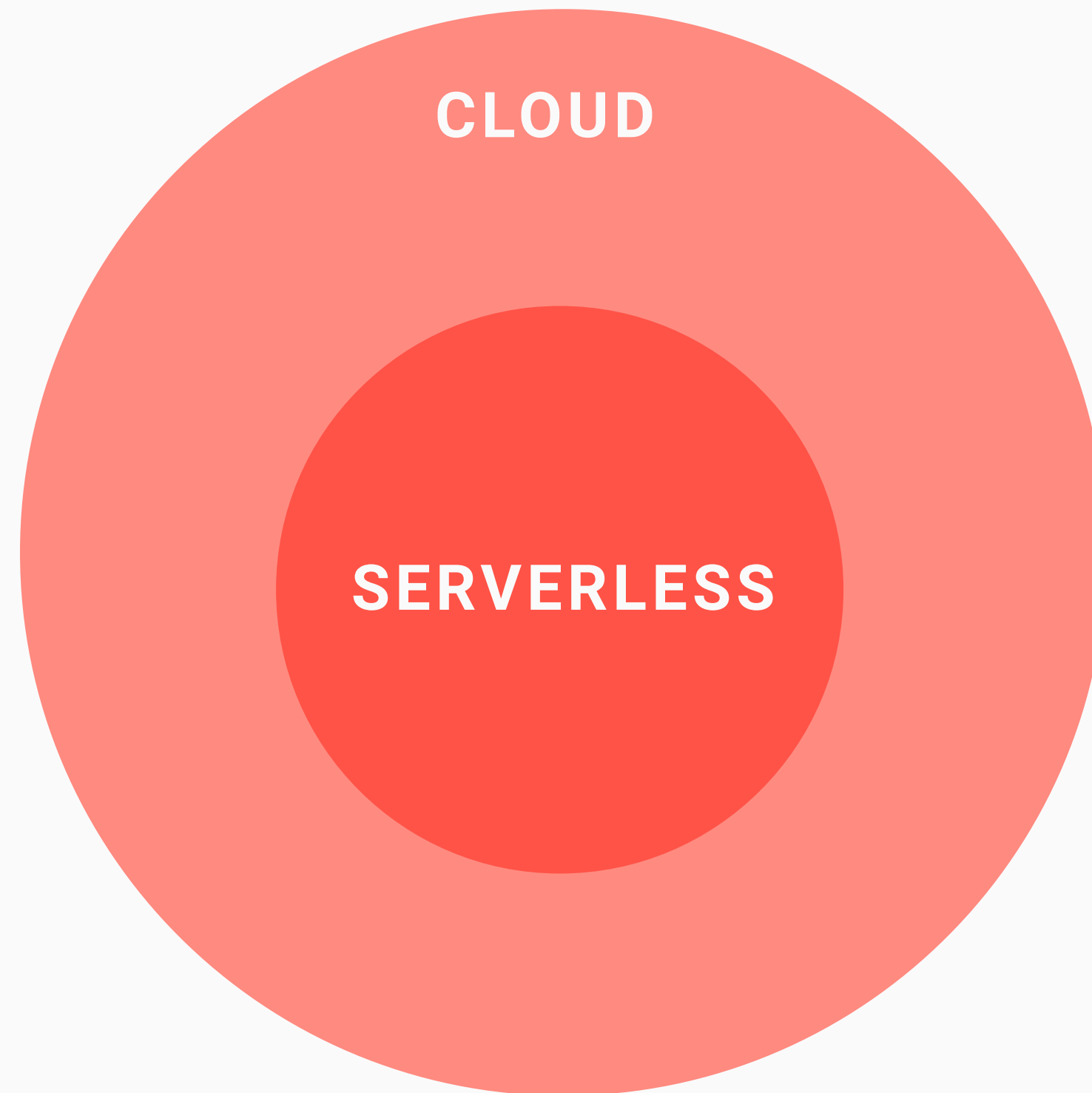


Serverless is

Zero overhead on hardware and maintenance

Yeah, That it.

Cloud and Serverless



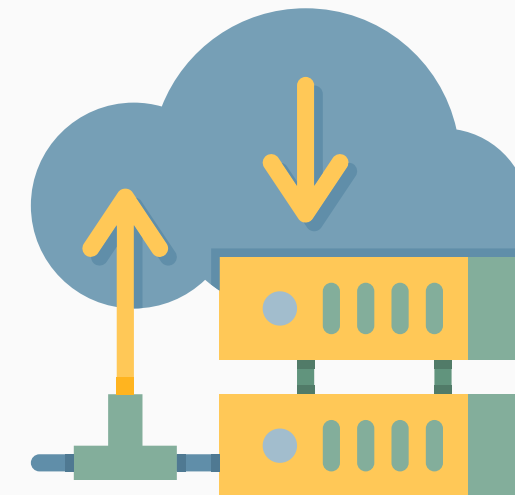
3 Main Architectures



ON-CLOUD



ON-PREMISE/LOCAL



HYBRID

Benefit of Cloud

ZERO CAPEX, OPEX-ONLY

No procurement cycle, no depreciation on hardware that loses value fast.

ELASTICITY

Burst to 500 concurrent users without buying for peak. On-prem you size for peak and waste the rest of the year.

NO OPERATION BURDEN

No driver/firmware maintenance, no power and cooling, no 3-year refresh.

CAPABILITY CEILING

This is the real differentiator, not cost. Frontier models you cannot self-host at any reasonable price.

SERVICE MANAGEMENT

managed adjacent services (object storage, identity, vector DB, monitoring).

TIME TO MARKET

Time to first prototype measured in hours.

Benefit of On-Premises

DATA NEVER CROSSES BOUNDARY

This collapses the hardest PDPA problems at once with sensitive-data consent

LEGAL COMPLY

Essential for validation and audit. Simplifies the legal position under data-protection law

LOW LATENCY

Deterministic disproportionately network for agentic pipelines

MARGINAL COST APPROACHES

This unlocks workloads that are economically impossible via API

FULL STACK CONTROL

Quantization scheme, guided decoding grammars, LoRA adapters for Thai medical terminology, custom sampling

OPERATES INDEPENDENTLY

Clinical continuity during internet outage.

Benefit of Hybrid

WORKLOAD ROUTING BY DATA CLASS

Sensitive data stays local while non-sensitive work uses the best available models

COMBINE LOAD SHARE

Local capacity sized for normal load, cloud absorbs peaks

BALANCES CAPITAL AND OPERATING EXPENDITURE

Manage each expenditure at your own pace

HYBRID OPERATIONAL

Cloud provides continuity if local infrastructure fails

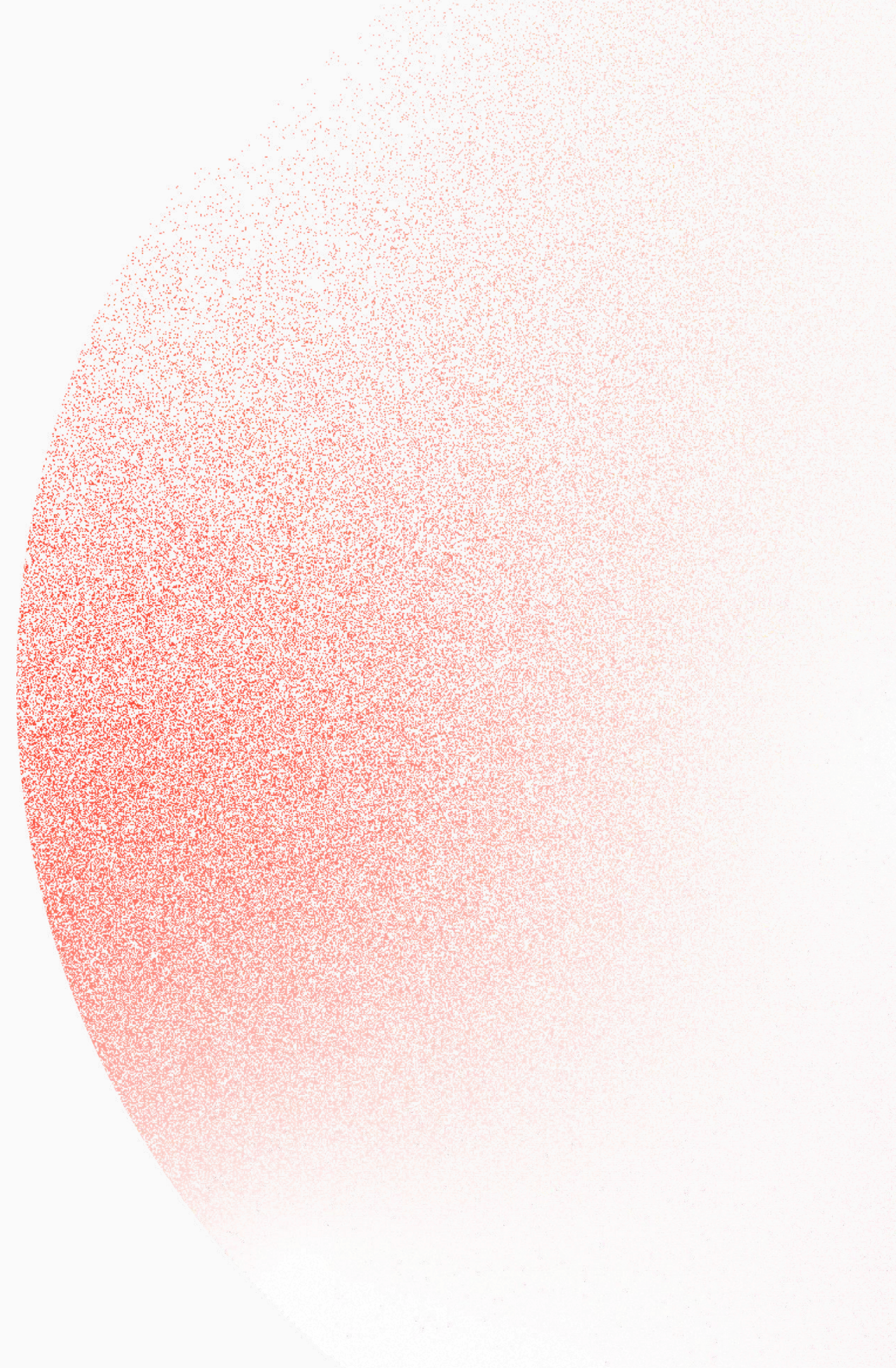
SERVICE TOLERANCE

Avoids dependence on a single provider or single site

GROWTH AS YOU GO

Allows incremental migration rather than an all-or-nothing commitment

Architecture Services



Load Balancer

Distributes incoming traffic across multiple servers to improve performance and availability.



Auto Scaling

Automatically adds or removes computing resources based on workload demand.



Network Gateway

Connects different networks and controls how data moves between them.



API Gateway

Acts as the entry point for APIs, routing requests to the correct backend service.

***APPLICATION PROGRAMMING INTERFACE**



Firewall

Monitors and filters network traffic to block unauthorized access and protect systems from threats.



Authentication

Verifies a user's or system's identity before allowing access.



CDN (Content Delivery Network)

Stores content on servers around the world so users can access it faster.



Object Storage

Stores files such as images, videos, documents, and backups.



Database

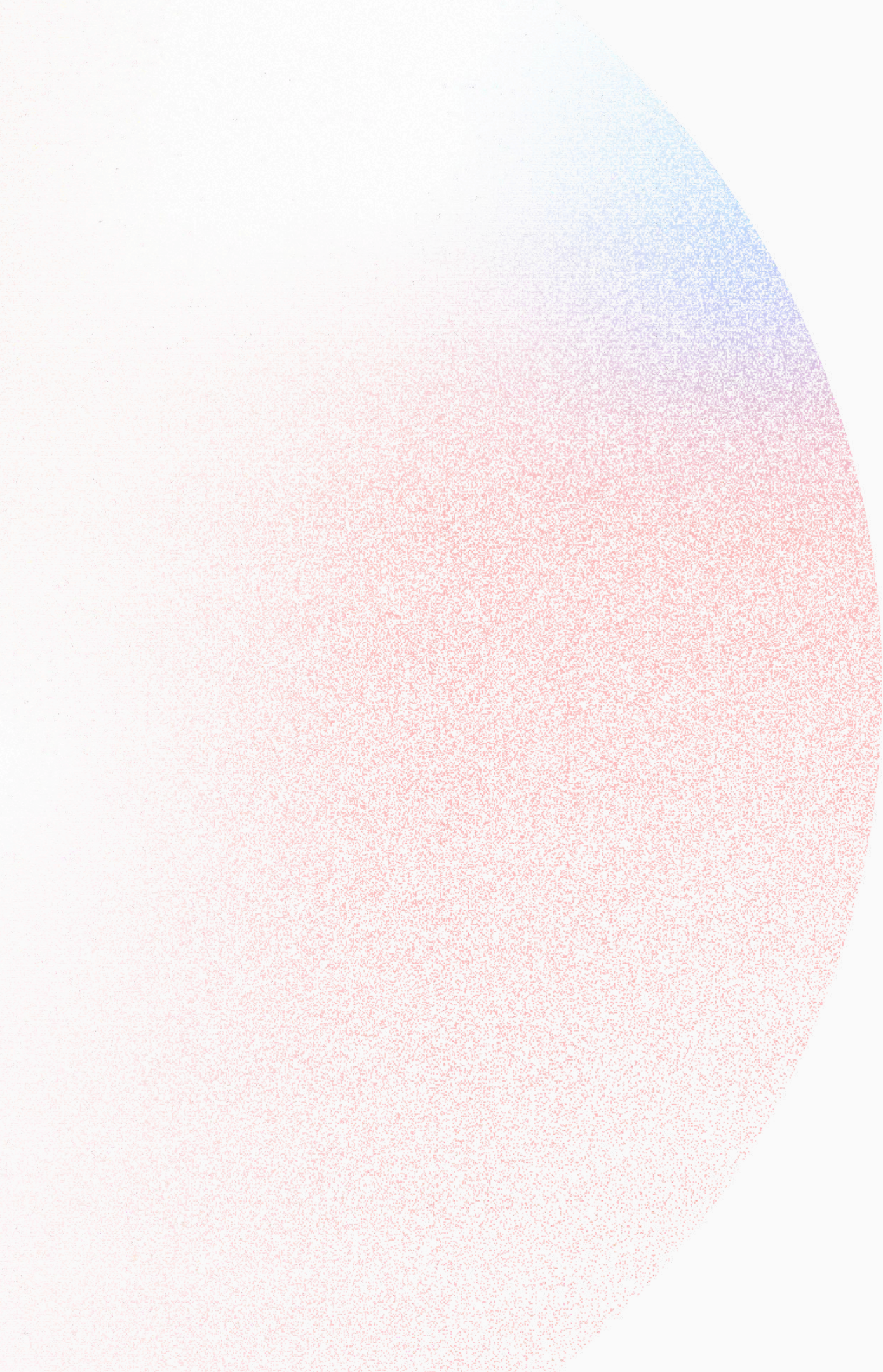
Stores and retrieves structured application data.



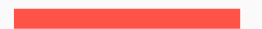
Cache (Meta Data)

Temporarily stores frequently used data to improve response time.





Workshop

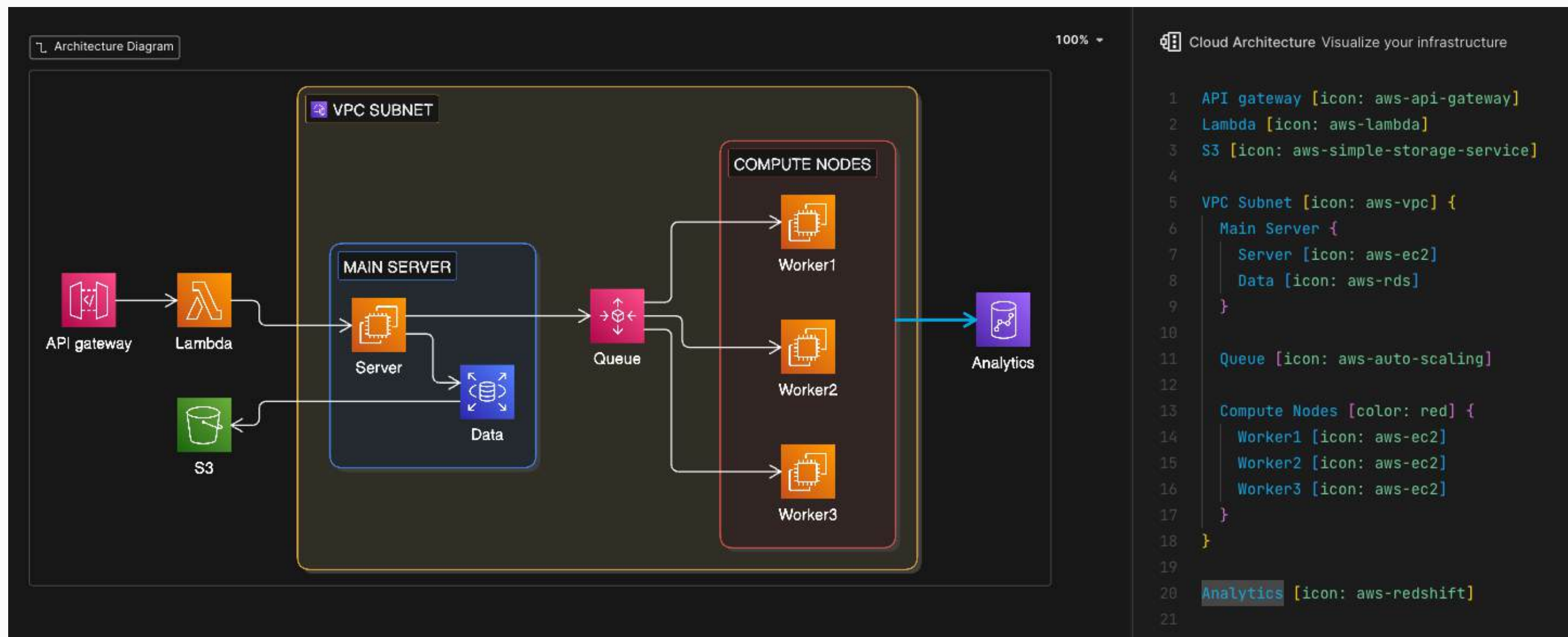


Eraser.io

DaaC = Diagram as a Code

<https://www.eraser.io/>

Diagram as Code is an approach for creating system architecture diagrams using text-based code rather than manually drawing them. This improves consistency, collaboration, reuse, and automation while reducing the risk of architecture documentation becoming outdated.



3 parts of DaaC

Object Node

Group Node

Link Node

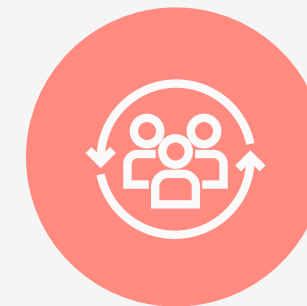
```
1
2 // Start Object
3 API gateway [icon: aws-api-gateway]
4 Lambda [icon: aws-lambda]
5 S3 [icon: aws-simple-storage-service]
6 // End Object
7
8 // Start Group
9 VPC Subnet [icon: aws-vpc] {
10     Main Server {
11         Server [icon: aws-ec2]
12         Data [icon: aws-rds]
13     }
14
15     Queue [icon: aws-auto-scaling]
16
17     Compute Nodes [color: red] {
18         Worker1 [icon: aws-ec2]
19         Worker2 [icon: aws-ec2]
20         Worker3 [icon: aws-ec2]
21     }
22 }
23 // End Group
24
25 // Start Link
26 API gateway > Lambda > Server > Data
27 Server > Queue
28 Queue > Worker1, Worker2, Worker3
29 S3 < Data
30 Compute Nodes > Analytics
31 // End Link
```


Exercises

Try to create your organization workflow
or any workflow you want



8
OBJECTS



2
GROUPS



4
LINKS

Group Exercises

Create an AI system for your organization that includes at least five architecture services, whether on-premises, on cloud, or hybrid.

Key Takeaway

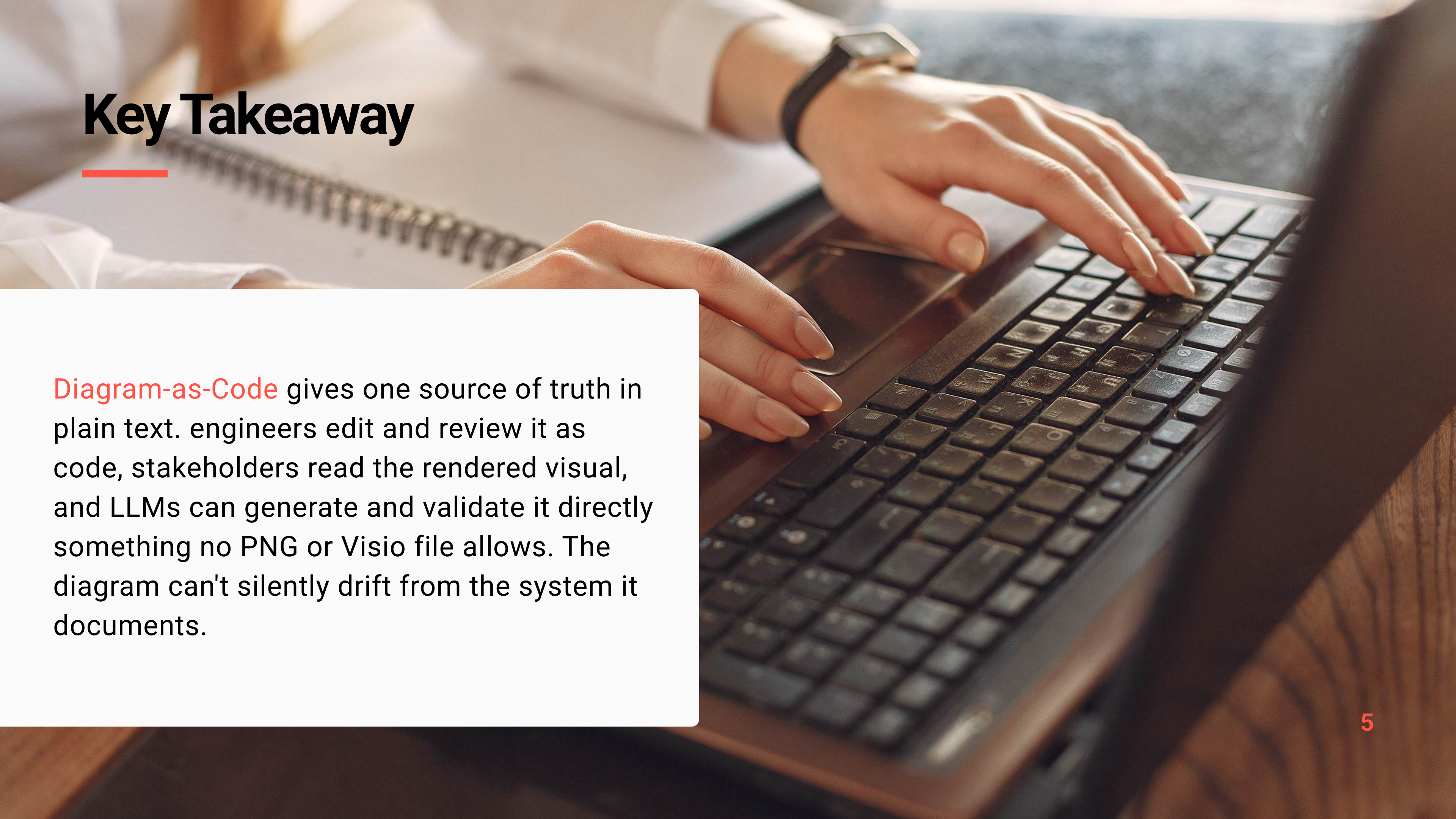
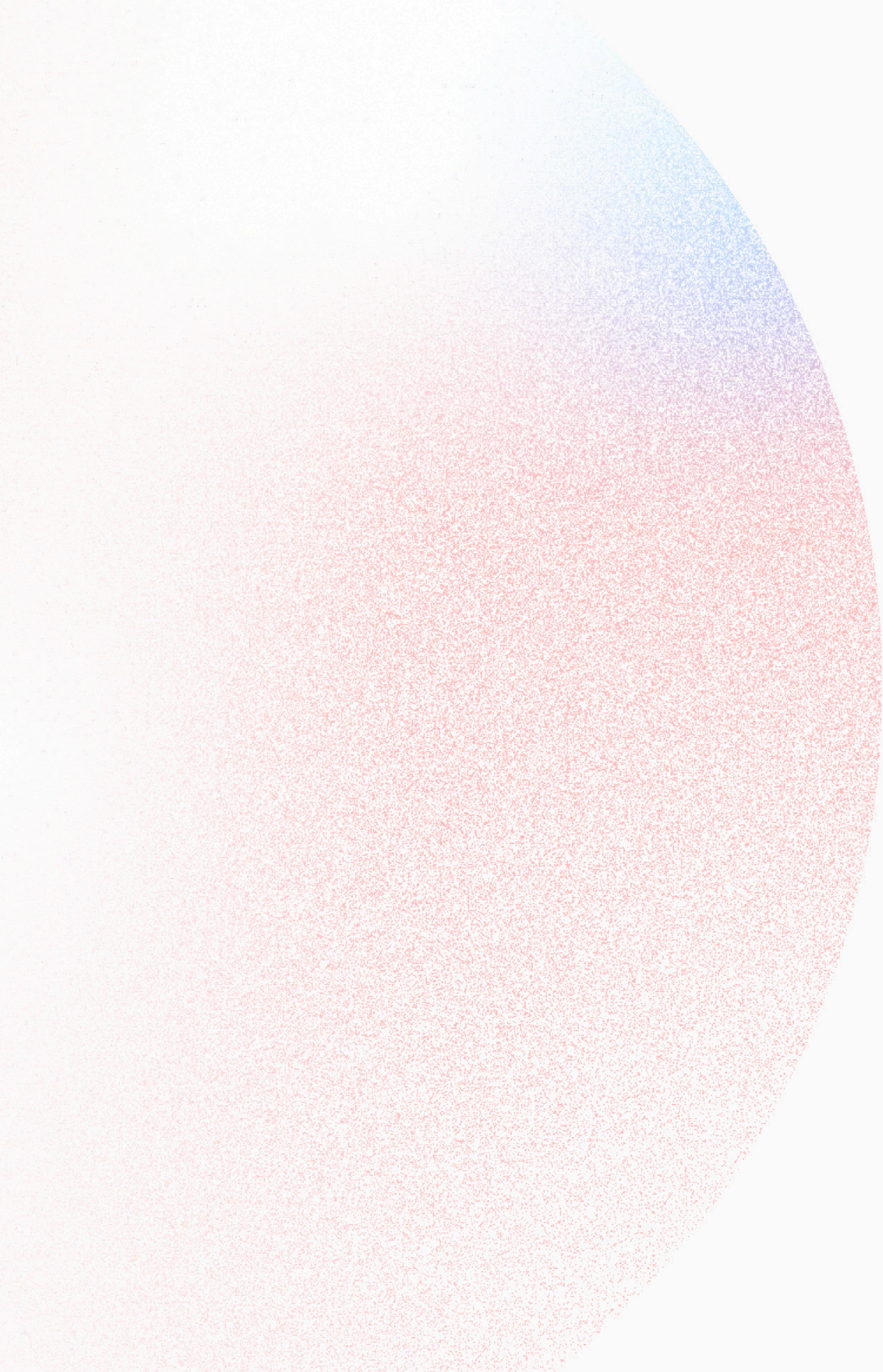


Diagram-as-Code gives one source of truth in plain text. engineers edit and review it as code, stakeholders read the rendered visual, and LLMs can generate and validate it directly something no PNG or Visio file allows. The diagram can't silently drift from the system it documents.



Thank You

